

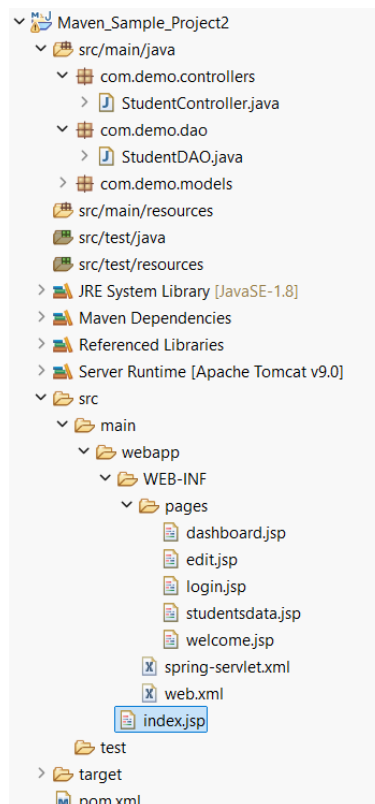
Spring MVC

Login Form Task :-

Steps to create Login Form Task Using Spring MVC :-

1. Create a Maven Web Project.
2. Add Spring MVC dependencies in pom.xml.
3. Create folder structure under WEB-INF/views.
4. Configure web.xml with DispatcherServlet.
5. Create dispatcher-servlet.xml for Spring MVC setup.
6. Enable component scan and view resolver.
7. Create Model class (optional).
8. Create DAO/Service to validate login credentials.
9. Create Controller with mappings for login page and login check.
10. Create login.jsp with email & password form.
11. Create dashboard.jsp and logout.jsp.
12. Deploy on Tomcat and test the login flow.

Project Structure –



Create the DAO (Data Access Object)/Repository Layer –

```
package com.demo.dao;

import java.util.List;

import org.springframework.dao.EmptyResultDataAccessException;
import org.springframework.jdbc.core.JdbcTemplate;

import com.demo.models.Student;

public class StudentDAO {

    JdbcTemplate template;

    public void setTemplate(JdbcTemplate template) {
        this.template = template;
    }

    public boolean registration(Student s) {
        // String sql="insert into student(full_name,email,password)
        // values('"+s.getFullname()+"','"+s.getEmail()+"','"+s.getPassword()+"','"+s.getAddress()+"')";
        // template.update(sql);

        String sql="insert into student(full_name,email,password,address) values(?,?,?,?)";
        template.update(sql,s.getFullname(),s.getEmail(),s.getPassword(),s.getAddress());
        return true;
    }

    public List<Student> getAllStudents(){
        String sql = "select * from student order by id DESC";

        return template.query(sql,(rs,rowNum)->{
            Student s=new Student();
            s.setId(rs.getInt("id"));
            s.setFullname(rs.getString("full_name"));
            s.setEmail(rs.getString("email"));
            s.setPassword(rs.getString("password"));
            s.setAddress(rs.getString("address"));
            return s;
        });
    }

    public void delete(int id) {
        String sql="delete from student where id=?";
        template.update(sql,id);
    }

    public Student getStudentById(int id) {
        String sql = "select * from student where id=?";
        try {
            return template.queryForObject(sql,(rs,numberofrows)->{
                Student s=new Student();
                s.setId(rs.getInt("id"));
                s.setFullname(rs.getString("full_name"));
                s.setEmail(rs.getString("email"));
                s.setPassword(rs.getString("password"));
                s.setAddress(rs.getString("address"));
                return s;
            });
        } catch (EmptyResultDataAccessException e) {
            return null;
        }
    }
}
```

```

    },id);
} catch (EmptyResultDataAccessException e) {
    return null;
}
}

public void updatestudent(Student s) {
String sql="update student set full_name=?,email=?,password=?,address=? where id=?";
template.update(sql,s.getFullname(),s.getEmail(),s.getPassword(),s.getAddress(),s.getId());
}

public Student login(String email, String password) {
String sql = "select * from student where email=? and password=?";
try {
return template.queryForObject(sql,(rs,numberofrows)->{
Student s=new Student();
s.setId(rs.getInt("id"));
s.setFullname(rs.getString("full_name"));
s.setEmail(rs.getString("email"));
s.setPassword(rs.getString("password"));
s.setAddress(rs.getString("address"));
return s;
},email,password);
} catch (EmptyResultDataAccessException e) {
return null;
}
}
}

```

Create the Controller Layer –

```
package com.demo.controllers;

import java.util.List;

import javax.servlet.http.HttpSession;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Controller;
import org.springframework.ui.Model;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.ModelAttribute;
import org.springframework.web.bind.annotation.PathVariable;
import org.springframework.web.bind.annotation.PostMapping;
import org.springframework.web.bind.annotation.RequestParam;

import com.demo.dao.StudentDAO;
import com.demo.models.Student;

@Controller
public class StudentController {

    // @PostMapping("/add")
    // public String addition(@RequestParam int num1, @RequestParam int num2,Model m)
    // {
    //     int result=num1+num2;
    //     System.out.println("Addition= "+result);
    //     m.addAttribute("result",result);
    //     return "welcome";    //welcome.jsp
    // }

    // @GetMapping("/welcome")
    // public String wel() {
    //     return "welcome";    //welcome.jsp
    // }

    @Autowired
    StudentDAO sdao;

    @PostMapping("/save")
    public String save(@ModelAttribute Student s) {
        boolean result=sdao.registration(s);
        if(result) {
            System.out.println("Registration Successfully Completed");
            return "redirect:/studentsdata";
        }else {
            System.out.println("Fail");
            return "redirect:/";
        }
    }
}
```

```

@GetMapping("/studentsdata")
public String studentsdata(Model m) {
    List<Student> list=sdao.getallstudents();
    m.addAttribute("list",list);
    return "studentsdata";    //studentsdata.jsp
}

@GetMapping("/deletestudent/{id}")
public String deletestudent(@PathVariable int id) {
    sdao.delete(id);
    return "redirect:/studentsdata";
}

@GetMapping("/profileedit/{id}")
public String profileedit(@PathVariable int id, Model m) {
    Student s= sdao.getStudentById(id);
    m.addAttribute("student",s);
    return "edit";    //edit.jsp
}

@PostMapping("/updatestudent")
public String updatestudent(@ModelAttribute Student s) {
    sdao.updatestudent(s);
    return "redirect:/studentsdata";
}

@GetMapping("/login")
public String login() {
    return "login";    //login.jsp
}

@PostMapping("/logincheck")
public String logincheck(@RequestParam String email,@RequestParam String password, HttpSession session, Model m) {
    Student s = sdao.login(email,password);

    if(s != null) {

        session.setAttribute("user",s);    //session start
        m.addAttribute("user",s);
        System.out.println("Login Successfully");
        return "redirect:/dashboard";
    }else {
        System.out.println("Login Fail");
        return "redirect:/login";
    }
}

@GetMapping("/dashboard")
public String dashboard(Model m, HttpSession session) {
    Student s = (Student) session.getAttribute("user");
    m.addAttribute("user",s);
    if(s == null) {
        return "redirect:/login";
    }
    return "dashboard";    //dashboard.jsp
}

@GetMapping("/logout")
public String logout(HttpSession session) {
    session.invalidate();    //session end
    return "redirect:/login";
}
}

```

login.jsp –

```
<%@ page language="java" contentType="text/html; charset=UTF-8"
    pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<title>Insert title here</title>
</head>
<body>
    <h2>Login Form</h2>
    <form action="logincheck" method="post">
        <input type="email" name="email" placeholder="Enter Email Id"><br><br>
        <input type="password" name="password" placeholder="Enter Password"><br><br>

        <button>Submit</button>
    </form>
</body>
</html>
```

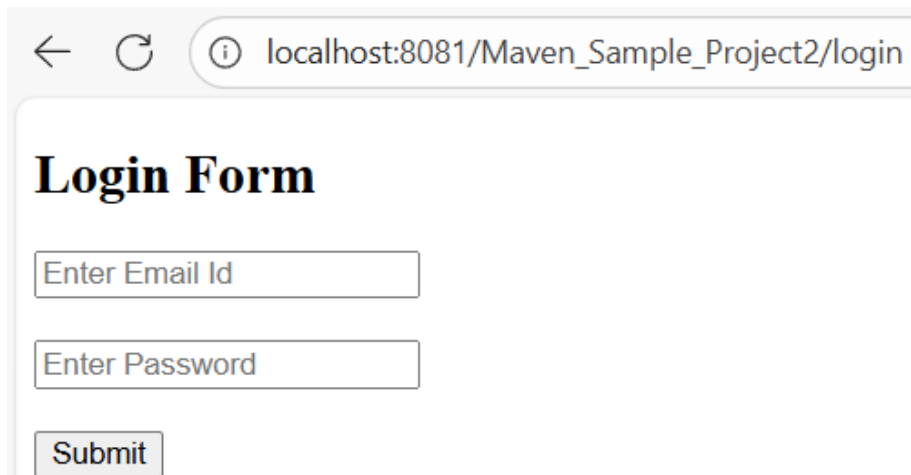
dashboard.jsp –

```
<%@ page language="java" contentType="text/html; charset=UTF-8"
    pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<title>Insert title here</title>
</head>
<body>
    <h1>Welcome to Dashboard</h1>

    ${user.fullname}<br>
    ${user.email}<br>
    ${user.address}<br>

    <a href="Logout">Logout</a>
</body>
</html>
```

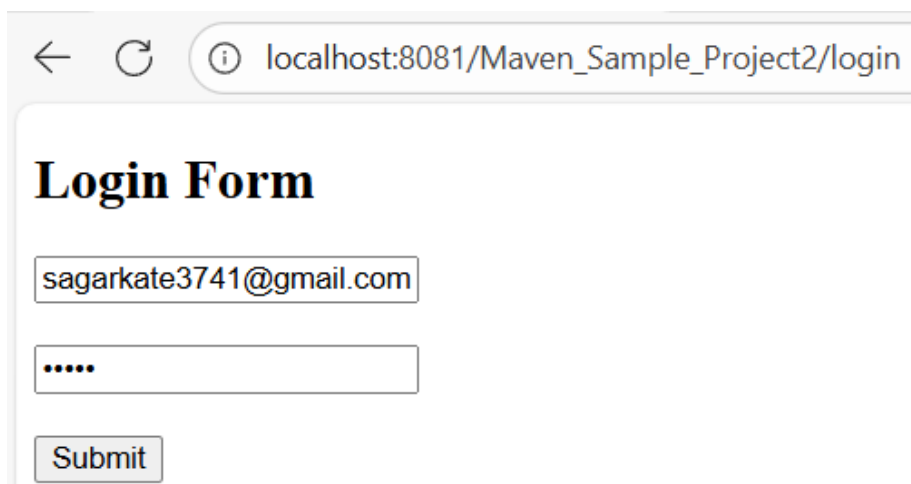
Test the Form Operations –



← ↻ ⓘ localhost:8081/Maven_Sample_Project2/login

Login Form

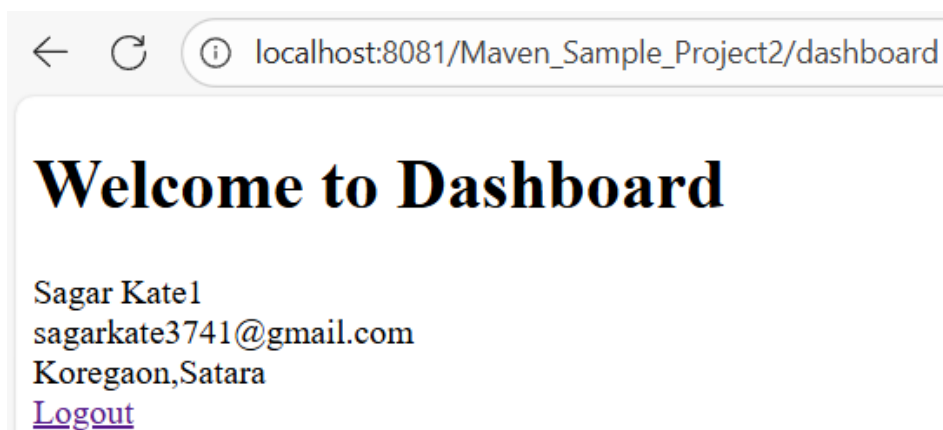
If the Email id & Password are correct then –



← ↻ ⓘ localhost:8081/Maven_Sample_Project2/login

Login Form

Output –

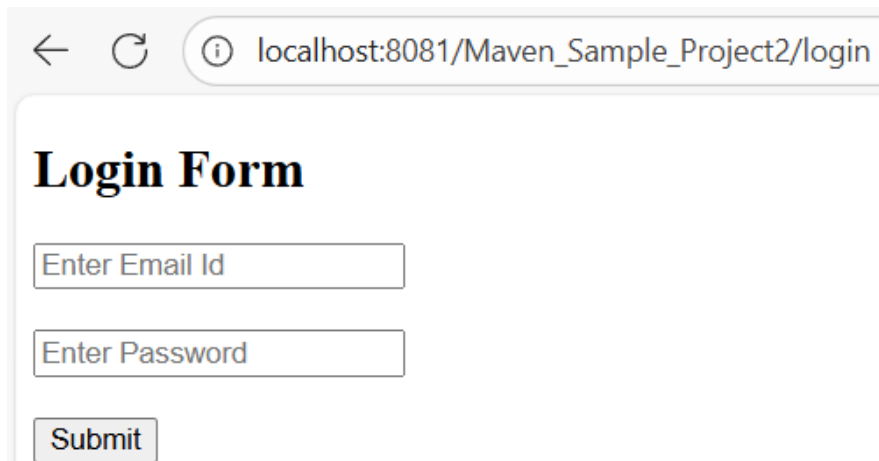


← ↻ ⓘ localhost:8081/Maven_Sample_Project2/dashboard

Welcome to Dashboard

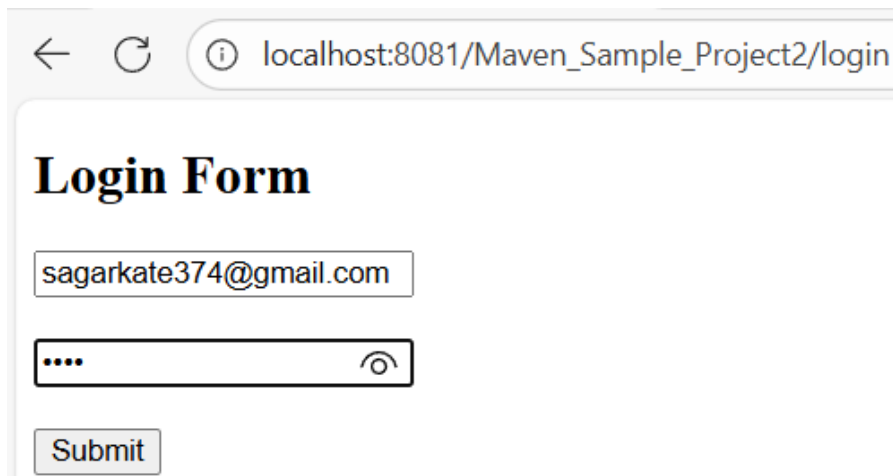
Sagar Kate1
sagarkate3741@gmail.com
Koregaon, Satara
[Logout](#)

After Logout –



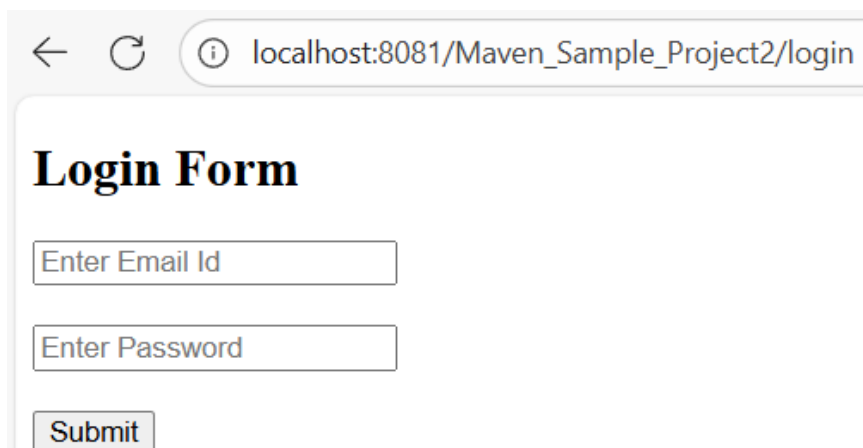
A screenshot of a web browser window. The address bar shows 'localhost:8081/Maven_Sample_Project2/login'. The page title is 'Login Form'. Below the title, there are two input fields: 'Enter Email Id' and 'Enter Password'. At the bottom, there is a 'Submit' button.

If the Email id & Password are incorrect then –



A screenshot of a web browser window. The address bar shows 'localhost:8081/Maven_Sample_Project2/login'. The page title is 'Login Form'. Below the title, the 'Enter Email Id' field contains the text 'sagarkate374@gmail.com'. The 'Enter Password' field contains four dots '....' and a toggle icon (an eye with a diagonal line through it). At the bottom, there is a 'Submit' button.

login.jsp -



A screenshot of a web browser window. The address bar shows 'localhost:8081/Maven_Sample_Project2/login'. The page title is 'Login Form'. Below the title, there are two input fields: 'Enter Email Id' and 'Enter Password'. At the bottom, there is a 'Submit' button.